
Privacy Analysis of Augmented Reality Glasses

12.23.2024 ~ 12.30.2025

Weekly Report 14

Michelle Jessica Bösigler
michellej.boesiger@unibas.ch

Department of Mathematics and Computer Science
University of Basel

30 December 2024

What has been done (Previous Week)

- Regarding the Bachelor Thesis: This is the link to my overleaf document - <https://www.overleaf.com/7936924994jytsxtynnbdc#9fefcf>
So far I have written:
 - Introduction - Frames the research objectives / context / challenges - basic and needs rework
 - Device Overview - Describes the glasses and their companion app - as this was part of the thesis description for understanding the functionality and privacy implications of the glasses.
 - Methodology and Experiments - Outlines the experimental setup, testbed configuration, and data collection methods - I expanded the chapter a lot more. Also whilst writing I refined the experiment a bit so that it is easier for analysis (especially for the voice commands). And I also introduced a little bit of automation where the voice commands are from a voice recording which helps with reproducibility, which I used for the recaptures performed. However, the chapter still needs a lot of improvements, for example in experiment design (right now its more like: what are the goals), as well as add more to Experimental procedure and other missing subsections.
 - Analysis of the Meta View app - An analysis which addresses questions 5, 6, 7 from the analysis goals, including trackers, Firebase integration, manifest components, and native libraries.
- Regarding Network Captures: I redid and uploaded the network captures. There were a few unexpected problems when capturing network traffic that resulted in a better understanding of my previous problems:
 - The POCO X5 5G phone does not allow Bluetooth and the App to be connected at the same time. What this means is that I am either able to connect to Bluetooth in Settings and be able to listen to music or connect to the App and transfer media, but not simultaneously.
 - The reason for this is because the phone is not compatible. Most online posts about the Meta glasses having a lot of problems are with phones are not officially supported. They may work (like the RedMi phone, but they will have their problems, technically the POCO phone "works" as well, only that there is a choice between the functionality you want in that moment).
 - And this is where I learned that the reason why I had so much issues with the RedMi phone in regards to the connection and the transfer of media captures is really just because the phone is not a supported model.

- I also tried with the Pixel 4 phone is not able to use the app because it does not have an acceptable security version on the phone.
 - The app also has VM detection which is why when I tried using an emulator previously (before I had a rooted phone and also for the dynamic analysis), it did not work properly.
 - But since I needed to redo the network captures, I got a supported phone (Samsung S24) and captured the network traffic there for the analysis. There were no problems in regards to connecting nor transferring media captures.
- Regarding Bluetooth Captures: I also tried to capture Bluetooth traffic with that phone, hoping it would be different.
 - Neither Micro:bits nor Adafruit was able to sniff the connection.
 - With the Ubertooth one I am able to follow the connection but I am not able to see my devices, and they are mostly all Malformed packets.
 - I had another sniffer, the nRF52840-Dongle. I wasn't able to get it to work previously so I stopped halfway. But now I managed to get it to work. This was able to sniff quite a bit of traffic and few were Malformed. Sadly none of the sources/destination were for the glasses.
- Regarding Decryption: It has been confirmed that the app uses SSL certificate pinning to detect and prevent MITM attacks, also identified through MobSF analysis. Initially, my focus was on files related to certificate handling, such as TrustManager, SSLContext, and X509TrustManager. Due to objection / Frida not being able to fully bypass certificate pinning, I looked deeper into custom certificates files including TrustManagerFactory using KeyStore and getAcceptedIssuers which suggest that the app very likely uses a custom certificate store as well for its SSL connections.

What will be done next week

- I will mainly only be doing the thesis, in regards to:
 - Results of the experiments
 - Analysis of the Network Traffic
 - This would mean that I should answer all the rest of the 6 questions by next week.
 - Methodology and Experiments - Improve on the Experimental procedure first. And if I finish answering the rest of the questions, then I can continue on the rest of the missing subsections.
- I did not have enough time to regenerate the images for the recaptured network traffic, so I will do them this week. I also thought of a few more images I can do, so I will also write the code and generate those images this week.

Meeting

Meeting canceled.